

Displacement reconstruction from measured accelerations and accuracy control of integration based on a low-frequency attenuation algorithm

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ABSTRACT

This study is aimed to reconstruct displacements from measured accelerations based on a low-frequency attenuation (LFA) algorithm. The integration of acceleration is conducted directly in frequency domain and the integral accuracy is governed by an accuracy control factor (ACF). A finite element simulation is conducted to compare the integral accuracy of this algorithm with those of several other conventional integration algorithms and then the superiority of the LFA algorithm is verified. At the same time, a numerical study and a vibration test rig with limit positions of impact are designed to investigate the influence of the ACF on integration accuracy for the LFA algorithm. Besides, three error evaluation indices are proposed to evaluate the integration accuracy. The investigations indicate that the determination of the value of the ACF should fully consider the level of low frequency noise, and it is better to select a relatively large value to ensure a good effect of eliminating the trend error and/or drift error. According to this research, the value of the ACF is suggested to be selected in the range from 0.90 to 0.98 so that integral results with high accuracy can be obtained in engineering applications.

1. Introduction

Accurately capturing vibration signals including dynamic displacement and velocity of structures has widely demands in various fields of engineering and science applications, such as the measurement of earthquakes [1–3], vibration test and control of structures [4–6], identification and condition monitoring of dynamic systems [7,8] and structural damage evaluation [9,10]. Besides, in transient finite element (FE) analysis, accurate displacement or velocity often should be exerted on structures as boundary conditions. Nowadays, there are two types of method to obtain dynamic displacements: a direct method and indirect one. The direct method captures dynamic displacements from measurements such as displacement transducers, geodetic sensors, photogrammetric techniques and radar measurements. The indirect method often obtains dynamic displacements through double numerical integration of accelerometer measurements. However, the direct method has pretty strict demands on test conditions such as the installation space and temperature which cannot be satisfied in many actual applications. Through double integration of acceleration signal measured by an

accelerometer to get displacement is relatively convenient and easy to realize, since there are diverse kinds of accelerometers and the acceleration can be measured conveniently with high accuracy in such cases. But the trend-term interference or drift error, which is usually blamed on the low frequency noise, often occurs in the process of integrating acceleration to displacement. It results in seriously paranoid displacement that cannot well meet engineering demands. Therefore, how to reduce the interferences of the trend-term error and drift error in acceleration integration is always what researchers greatly care about.

So far, studies on algorithms for displacement reconstruction from measured acceleration mainly include the time domain integration approach [11–16], the digital filter approach [17–19] and the frequency domain integration approach [20–22]. Chen et al. [16] proposed a method that the integral displacement waveform is fitted through a polynomial and then the fitted polynomial is subtracted to remove the trend-term. This method can well reduce the error caused by initial velocity and displacement as well as the direct-current (DC) error of accelerometers, but the influence of low-frequency noise on integration accuracy is hard to control and thus this method doesn't have a very good engineering application. Digital filters including the finite impulse

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FE	finite element
FIR	finite impulse response
IIR	infinite impulse response
DC	direct-current
LFA	low-frequency attenuation
ACF	accuracy control factor
FFT	fast Fourier transform
IFFT	inverse fast Fourier transform
PF	polynomial fitting
FDI	frequency domain integration
LFCO	low frequency cut-off
Erp	average peak error
Err	average deviation error
Ers	absolute value error

response(FIR) filters and infinite impulse response (IIR) filters are broadly used to reconstruct displacement from measured acceleration [17–19]. The digital filter methods can well suppress the low frequency noise but phase errors are caused as well [22]. In Ref. [23,24], the low-frequency cut-off (LFCO) algorithm is employed for displacement reconstruction from measured acceleration. The LFCO algorithm starts with making fast Fourier transform (FFT) on acceleration signals and then the low-frequency signals below the cut-off frequency are directly set to zero. Later, an inverse fast Fourier transform (IFFT) is conducted to reconstruct displacement. This algorithm can completely eliminate the low-frequency noise below the cut-off frequency selected, but the effective low-frequency signal is also removed and the selection of the cut-off frequency is strongly subjective and empirical [13]. Lee et al. [25] presented a new FIR filter on the basis of a minimization problem of a governing equation derived from an inverse problem. Besides, a regularization factor is introduced to govern the integration accuracy.

All in all, at present, a large number of studies have been conducted on displacement reconstructions from measured acceleration data and several integration methods are proposed. However, the drift errors or trend term errors are usually reduced through abandoning the low-frequency signals for most of these models. The method proposed in Ref. [25] has certain advancement since it can not only effectively attenuate the low-frequency noise, but also can avoid the loss of useful low-frequency signals. But the new FIR filter in Refs. [25] is complexly designed and its transition zone becomes shorter that oscillations will occur in the frequency domain when the target frequency is low.

This paper presented an improved low-frequency attenuation (LFA) algorithm, where the integration process is simplified and oscillations in the frequency domain are avoided since the displacement is directly reconstructed in frequency domain and the filter design is omitted. And then a FE simulation study is conducted to compare the integral results of the LFA algorithm and those of other conventional integration algorithms. At the same time, a numerical study and a vibration test rig are designed to investigate the influence of the accuracy control factor (ACF) on integral accuracy and experimental validation of the LFA algorithm. Besides, three error evaluation indices are proposed to evaluate the integration result. Finally, several conclusions are derived from the comparison and the selection strategy for the ACF is given for engineering applications. The LFA algorithm has been encoded into engineering software for signal analysis and processing.

2. Integration algorithm

When the measured acceleration signal is employed to reconstruct displacement, there are many types of error sources that affect the integration accuracy. For example, aliasing phenomenon, which occurs

due to discrete-time samples, can introduce low-frequency errors in the acceleration waveforms [21]. Besides, instabilities of sampling rates would introduce additional noise as well [13]. What's more, low frequency noises would be amplified through integration and propagate with time. Hence, the key to assuring good integration accuracy and avoiding drift errors is trying to eliminate the low-frequency noise component of the measured acceleration as much as possible. In this section, we will talk about several conventional integration algorithms and the LFA algorithm.

2.1. Polynomial fitting algorithm

The polynomial fitting (PF) algorithm is a typical time domain integration method which can meet the requirements of real-time computation, and has small amount of calculation and can also avoid the truncation error caused by the Fourier transform. Hence, it is widely applied in engineering test [11–14].

A measured acceleration signal $a(t)$ usually contains a DC error δ caused by various interference factors:

$$a(t) = f(t) + \delta \quad (1)$$

where $f(t)$ is the real acceleration. The velocity obtained by integrating the acceleration waveform is

$$v(t) = \int a(t)dt + v_0 = \int (f(t) + \delta)dt + v_0 = \int f(t)dt + \delta t + \varepsilon + v_0 \quad (2)$$

where v_0 is the real initial velocity and ε is an integration constant that can be determined by the initial value of $v(t)$. The displacement obtained by integrating the velocity waveform is

$$\begin{aligned} s(t) &= \int v(t)dt + s_0 = \int \left(\int f(t)dt + \delta t + \varepsilon + v_0 \right) dt + s_0 \\ &= \int \left(\int f(t)dt \right) dt + 0.5\delta t^2 + \varepsilon t + v_0 t + s_0 \end{aligned} \quad (3)$$

where s_0 is the real initial displacement and ε is the integration constant that can be determined by the initial value of $s(t)$.

Obviously, the DC error δ is gradually amplified during the integration processes. Because the initial velocity and displacement are unknown, the integral velocity has a linear error item $\delta t + \varepsilon$, and thus the integral displacement has a quadratic error item $0.5\delta t^2 + \varepsilon t + \varepsilon$. The linear error item and quadratic error item can be eliminated through the PF algorithm.

Through numerical integration methods such as the trapezoid formula, Simpson formula, Newton-Cotes formula, the compound trapezoid formula, the central difference method, the Lagrange Interpolation, etc., the velocity and displacement can be reconstructed from the measured acceleration waveform and expressed as below.

$$\begin{cases} v = (t_i, v_i), & (i = 0, 1, \dots, n-1) \\ s = (t_i, s_i), & (i = 0, 1, \dots, n-1) \end{cases} \quad (4)$$

Assume that there is a polynomial $f_m(t)$ that assures the minimum quadratic sum of the difference between $f_m(t)$ and (t_i, v_i) :

$$I = \min \sum_{i=0}^{n-1} [v_i - f_m(t_i)]^2 \quad (5)$$

where

$$f_m(t) = \sum_{k=0}^m p_k t^k \quad (6)$$

and $p_k (k = 1, 2, \dots, m)$ are coefficients of the polynomial [16].

According to Eq. (5), if we can find a set of coefficients $p_k (k = 1, 2, \dots, m)$ that ensure the minimum value of I , the trend term error can be determined. The governing equation (5) should satisfy that the

partial derivative of I with regard to $p_j (j = 1, 2, \dots, m)$ is equal to zero [16]:

$$\frac{\partial I}{\partial p_j} = 2 \sum_{i=0}^{n-1} \left[v_i - \sum_{k=0}^m p_k (t_i^k) \right] t_i^j = 0, \quad (j=0, 1, \dots, m) \quad (7)$$

According to Eq. (7), one can derive the following equation:

$$\sum_{k=0}^m \left(\sum_{i=0}^{n-1} t_i^{j+k} \right) p_k = \sum_{i=0}^{n-1} t_i^j v_i \quad (8)$$

Rewriting Eq. (8) in a matrix form [16].

$$\begin{bmatrix} n & \sum_{i=0}^{n-1} t_i & \dots & \sum_{i=0}^{n-1} t_i^m \\ \sum_{i=0}^{n-1} t_i & \sum_{i=0}^{n-1} t_i^2 & \dots & \sum_{i=0}^{n-1} t_i^{m+1} \\ \dots & \dots & \dots & \dots \\ \sum_{i=0}^{n-1} t_i^m & \sum_{i=0}^{n-1} t_i^{m+1} & \dots & \sum_{i=0}^{n-1} t_i^{2m} \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ \dots \\ p_m \end{bmatrix} = \begin{bmatrix} \sum_{i=0}^{n-1} v_i \\ \sum_{i=0}^{n-1} t_i v_i \\ \dots \\ \sum_{i=0}^{n-1} t_i^m v_i \end{bmatrix} \quad (9)$$

It can be validated that the coefficient matrix of Eq. (9) is a positive definite so that Eq. (9) has a single solution. Once the coefficients $p_k (k = 1, 2)$ are determined, the trend term error $f_m(t)$ is derived; and then the real displacement $s(t)$ can be obtained through Eq. (10).

$$s(t) = \iint a(t) dt dt - f_2(t) = \iint a(t) dt dt - \sum_{k=0}^2 p_k t^k \quad (10)$$

Through the summary and deduction of the PF algorithm talked above, the flowchart of the PF algorithm is drawn in Fig. 1:

2.2. Filter algorithm

The so-called trend term generally refers to signal components whose periods are greater than record lengths of test data, namely the linear term or those slowly changed nonlinear term.

$$a_m(t) = a(t) + a_e(t) = a(t) + a_n(t) + c \quad (11)$$

where $a_m(t)$ is the measured acceleration at a certain time t , $a(t)$ is the real acceleration, $a_e(t)$ is the test error of accelerator, c is the DC component of the measured acceleration, $a_n(t)$ is the random noise.

The velocity can be obtained by integrating the measured acceleration signal.

$$v(t) = \int a_m(t) dt = \int a(t) dt + \int a_e(t) dt = \int a(t) dt + \int a_n(t) dt + ct + D \quad (12)$$

The integrated velocity contains a noise component $\int a_n(t) dt$, the linear trend error ct which is caused by zero drift of accelerator, and the constant error term D . If these error terms cannot be identified and eliminated, they will be further amplified when integrating velocity to displacement and thus wrong displacements will be obtained.

The PF approach can effectively suppress or eliminate the influence of the linear trend term error ct and the constant error term D . However,

it cannot well reduce the interference of the random noise component $\int a_n(t) dt$ since it's based on the assumption that the test error of accelerator is a constant.

Because the low frequency noise of the acceleration signal will be amplified during the process of displacement reconstruction, added with that the accelerator generally has bad test accuracy at low frequency span (especially below 10 Hz), the signals with low frequencies are the main source of integral error. In many engineering applications, the high-pass filter or band-pass filter method is an effective way to reduce integral error [17–19].

Digital filter can be expressed in the frequency domain:

$$y(r) = \sum_{k=0}^{N-1} H(k) X(k) e^{j2\pi k r / N} \quad (13)$$

where $y(r)$, ($r = 0, 1, \dots, N-1$) is the filtered time series of the input signal; $X(k)$ is the Fourier transform of the input time series $x(r)$ and $H(k)$ is the transfer function for an ideal high-pass filter:

$$H(k) = \begin{cases} 0, & (k\Delta f < f_d) \\ 1, & (k\Delta f \geq f_d) \end{cases} \quad (14)$$

where f_d is the cut-off frequency; Δf is the frequency resolution.

A real high-pass filter consists of a capacitor and resistor which are denoted by C and R , respectively, and its transfer function is given as below:

$$\begin{cases} |H(f)| = \frac{f/f_d}{\sqrt{1 + (f/f_d)^2}} \\ \varphi(f) = 90^\circ - \tan^{-1} \left(\frac{f}{f_d} \right), f_d = \frac{1}{2\pi RC} \end{cases} \quad (15)$$

The amplitude-frequency characteristics of the real and ideal high-pass filter are drawn in Fig. 2. The high-pass filter can filter the low frequency noise with a proper selection of the cut-off frequency and remove the zero drift of the accelerometer. The flowchart of the high-pass filter algorithm for displacement reconstruction is shown in Fig. 3.

2.3. Frequency domain integration method

Because of the transition zone of a real filter, the low frequency noise signal cannot be effectively eliminated. What's more, the filter method is easy to cause phase error. The frequency domain integration (FDI) method uses the Fourier transform and inverse Fourier transform as well as the transfer function to reconstruct displacement from test acceleration. The integration is substituted by Fourier transform, and sine and cosine transforms are directly implemented in frequency domain so that accumulation and amplification of small errors of integration in time domain is avoided. But choice of an amplitude control function is a difficulty for the FDI algorithm.

For a discrete acceleration series $a(n)$ with a length of N in the time domain, a FFT is made:

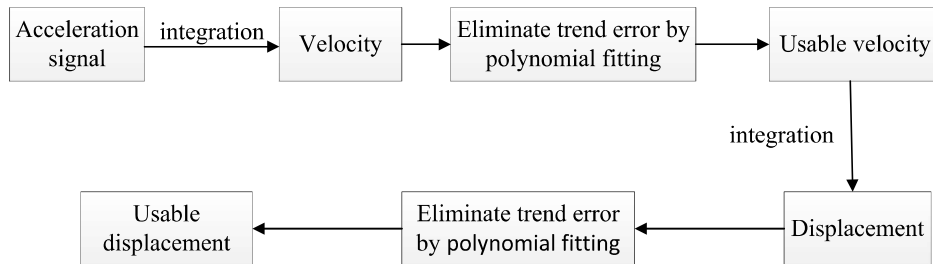


Fig. 1. Flowchart of the PF method.

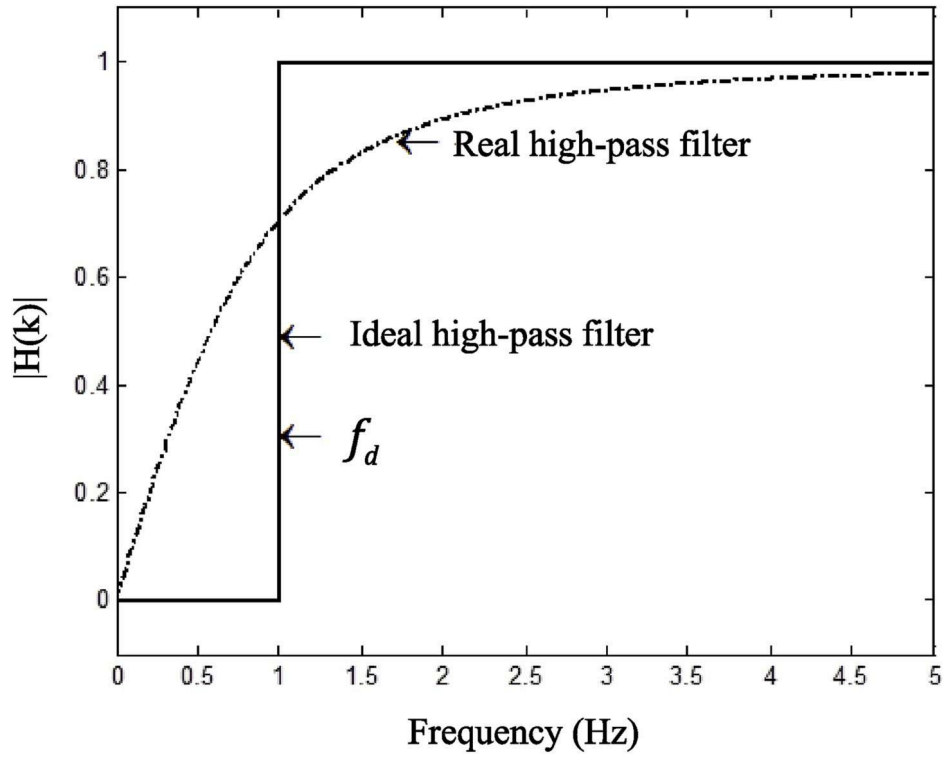


Fig. 2. Amplitude-frequency characteristics for a high-pass filter.

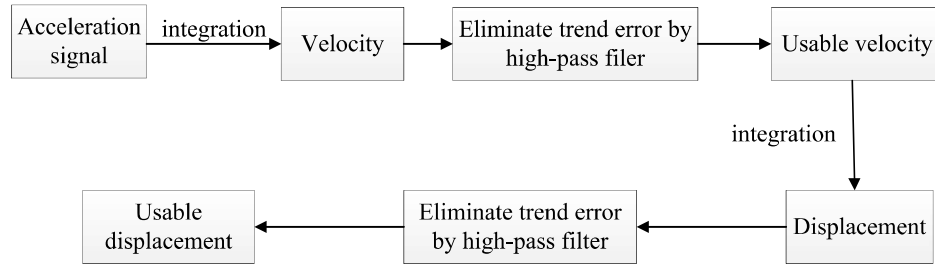


Fig. 3. Flowchart of high-pass filter method to reconstruct displacement.

$$\begin{cases} a(k) = \frac{2}{N} \sum_{n=1}^N a(n) e^{\left[\frac{-2\pi i(k-1)(n-1)}{N} \right]} = a_k + b_k i \\ f(k) = \frac{(n-1)f_0}{2N}, 1 \leq k \leq N \end{cases} \quad (16)$$

where f_0 is the sample frequency, $a(k)$ is the complex sequence of $a(n)$ in the frequency domain after the Fourier transform and $f(k)$ is the corresponding frequency.

Rewriting $a(k)$ in a simple harmonic expression:

$$\begin{cases} a(t)_k = A_k \cos(\omega_k t + \varphi_k) \\ A_k = \sqrt{a_k^2 + b_k^2} \\ \omega_k = 2\pi f_k \\ \varphi_k = \arctan(b_k/a_k) \end{cases} \quad (17)$$

where A_k , ω_k and φ_k , are amplitude, circular frequency and phase angle of the harmonic waveform, respectively. By twice integration of Eq. (17) in the frequency domain, the harmonic displacement for each frequency component is

$$\begin{cases} s(t)_k = A_{sk} \cos(\omega_{sk} t + \varphi_{sk}) \\ A_{sk} = A_k / \omega_k^2, \varphi_{sk} = \varphi_k - \pi, \omega_{sk} = \omega_k \end{cases} \quad (18)$$

According to the principle of signal superposition, the displacement can be reconstructed by

$$s(t) = \sum_{k=1}^N s(t)_k = \sum_{k=1}^N A_{sk} \cos(\omega_{sk} t + \varphi_{sk}) \quad (19)$$

Hence, the frequency integration approach talked above can be explained through a Fourier transform and an inverse Fourier transform as described by

$$\begin{cases} F(s(t)) = H_E(\omega) F(\bar{a}(t)) = -\frac{1}{\omega^2} F(\bar{a}(t)) \\ s(t) = -F^{-1} \left[\frac{1}{\omega^2} F(\bar{a}(t)) \right] \end{cases} \quad (20)$$

where $H_E(\omega)$ is the transfer function of $s(t)$ with respect to $\bar{a}(t)$, with $\bar{a}(t)$ being the measured acceleration and $s(t)$ being the reconstructed displacement.

According to Eq. (20), the low frequency component of the measured acceleration would be amplified while the high frequency component

would be attenuated during integration. Hence, the key to reducing integration error for displacement is minimizing the low-frequency noises. To reduce the trend term error or drift error caused by low-frequency noises, an accuracy function $\varphi(\omega)$ is usually introduced in Eq. (20) to govern the integration accuracy of the FDI algorithm as described by

$$s(t) = -F^{-1} \left[\frac{1}{\omega^2} \varphi(\omega) F(\bar{a}(t)) \right] \quad (21)$$

2.3.1. The low frequency cut-off integration algorithm

At present, the low frequency cut-off (LFCO) algorithm is one of the commonly used FDI algorithms for displacement reconstruction. The LFCO algorithm needs to select a cut-off frequency f_T . Suppose the lowest main frequency of the acceleration signal is f_s , the cut-off frequency should meet that $f_T \leq f_s$. In the whole frequency domain, the accuracy control function $\varphi(\omega)$ is defined by

$$\begin{cases} \varphi(\omega) = 0, & \omega < 2\pi f_T \\ \varphi(\omega) = 1, & \omega \geq 2\pi f_T \end{cases} \quad (22)$$

The accuracy function and the amplitude-frequency characteristics of s and $\bar{a}$ for the LFCO algorithm are drawn in Figs. 4 and 5, respectively.

From Figs. 4 and 5, through the accuracy control function, the LFCO algorithm directly set the amplitude of the acceleration signal whose frequency is below the cut-off frequency to be zero so that the low frequency noise is eliminated. However, the effective low frequency signals are also removed. The flowchart of the LFCO algorithm is drawn in Fig. 6.

2.3.2. The LFA algorithm

The LFA algorithm initiates from the governing equation for displacement reconstruction from measured accelerations, which is derived based on an inverse problem [20,25].

$$\min \Pi(s) = \frac{\beta^2}{2} \int_{T_1}^{T_2} s^2 dt + \frac{1}{2} \int_{T_1}^{T_2} \left(\frac{d^2 s}{dt^2} - \bar{a} \right)^2 dt \quad (23)$$

where the first term is a regularized equation which is used to eliminate the ill posedness and β is the regularization factor, and the second term is

employed to control the trend term error of measured acceleration.

Through solving the double objective optimization problem of Eq. (23), the transfer function $H_B(f)$ of the output displacement s and input acceleration $\bar{a}$ is derived [25].

$$H_B(f) = -\frac{\omega^2}{\omega^4 + \beta^2} = -\frac{(2\pi f)^2}{(2\pi f)^4 + \beta^2} \quad (24)$$

where ω is the circular frequency. Let $\hat{a} = \frac{d^2 s}{dt^2}$ is the output acceleration, the transfer function of the output $\hat{a}$ and input acceleration $\bar{a}$ is

$$H_B^{acc}(f) = \frac{\omega^4}{\omega^4 + \beta^2} = \frac{(2\pi f)^4}{(2\pi f)^4 + \beta^2} \quad (25)$$

A target frequency f_T is introduced according to the base frequency of the measured acceleration. And then Eqs. (24) and (25) are normalized by

$$\begin{cases} H_B(\tilde{f}) = -\frac{\tilde{f}^2}{[\tilde{f}^4 + \beta^2 / (2\pi f_T)^4]} \\ H_B^{acc}(\tilde{f}) = \frac{\omega^4}{\omega^4 + \beta^2} = \frac{\tilde{f}^4}{[\tilde{f}^4 + \beta^2 / (2\pi f_T)^4]} \end{cases} \quad (26)$$

where $\tilde{f}$ is the normalized frequency and $\tilde{f} = f/f_T$. An accuracy control parameter α_T is defined as below

$$\alpha_T = |H_B(\tilde{f})|_{\tilde{f}=1} = \frac{1}{1 + \beta^2 / (2\pi f_T)^4}, 0 < \alpha_T \leq 1 \quad (27)$$

The regularization factor can be derived according to Eq. (27)

$$\beta^2 = \frac{(2\pi f_T)^4}{\alpha_T} - (2\pi f_T)^4 \quad (28)$$

Substituting Eq. (28) into Eq. (26) yields the normalized amplitude-frequency characteristic of s and $\bar{a}$ and that of $\hat{a}$ and $\bar{a}$.

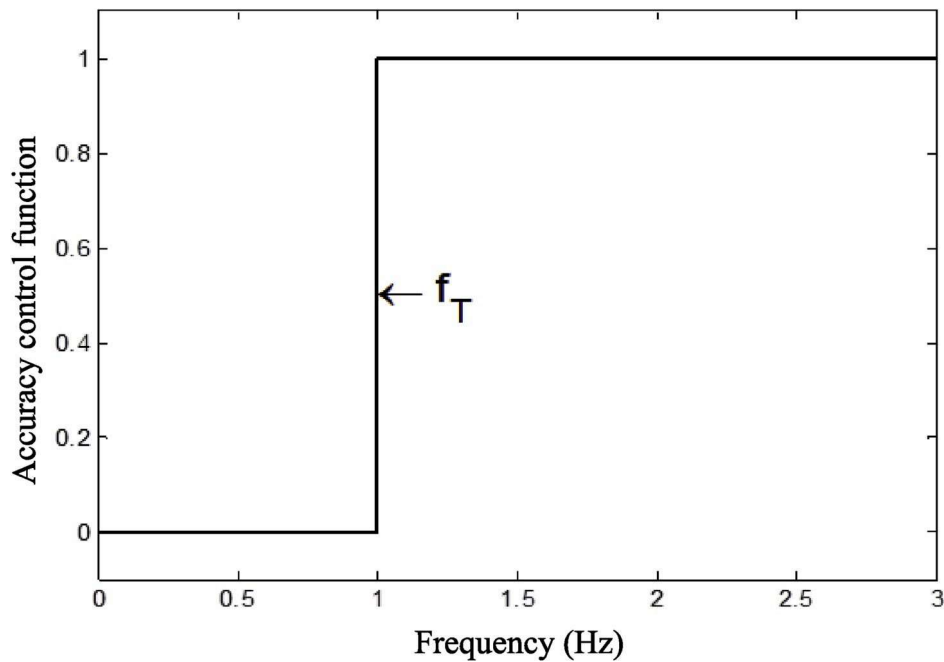


Fig. 4. The accuracy control function for the LFCO algorithm.

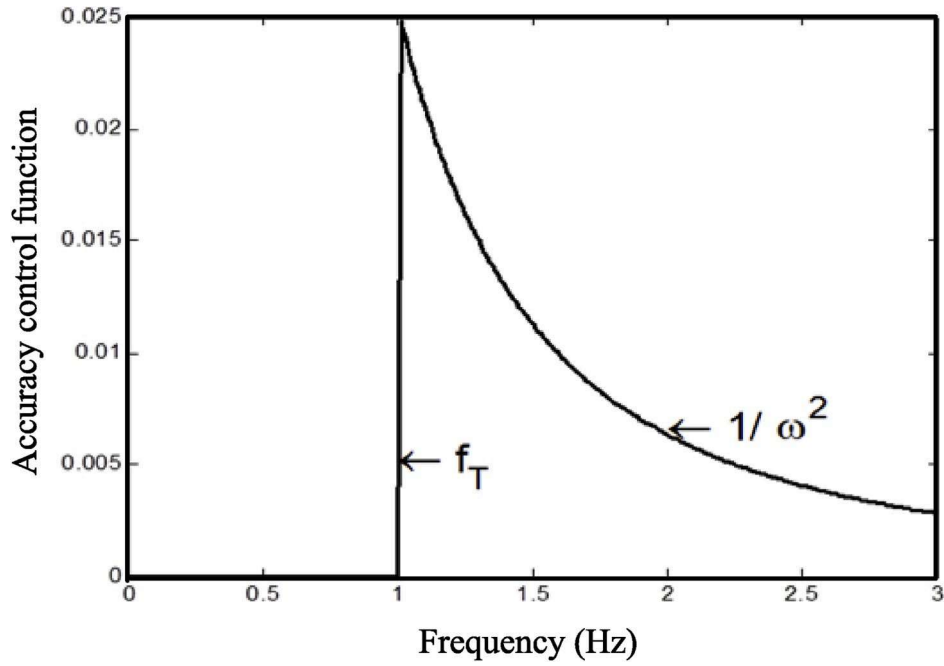


Fig. 5. Amplitude-frequency characteristics of s and $\bar{a}$ for the LFCO algorithm.

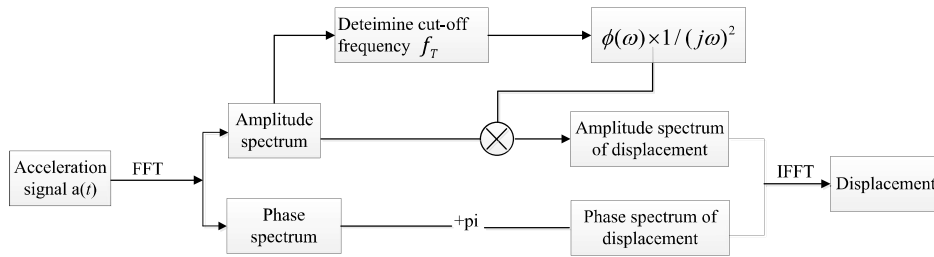


Fig. 6. Flowchart of the LFCO algorithm.

$$\begin{cases} H_B(\tilde{f}) = -\frac{\tilde{f}^2}{(\tilde{f}^4 - 1 + \frac{1}{\alpha_T})} \\ H_B^{acc}(\tilde{f}) = \frac{\tilde{f}^4}{(\tilde{f}^4 - 1 + \frac{1}{\alpha_T})} \end{cases} \quad (29)$$

Based on the amplitude-frequency characteristics of the output and input signals, the LFA algorithm is then proposed and the flowchart of this integration algorithm is presented in Fig. 7. The acceleration signal is directly integrated in frequency domain to reconstruct displacement and the ACF α_T is employed to assure the integration accuracy.

The amplitude-frequency characteristics of $\hat{a}$ and $\bar{a}$ with different values of ACF α_T are drawn in Fig. 8. It is shown that the LFA method rapidly suppresses the frequency far below the target frequency f_T , but the signals with frequencies around or above the target frequency are effectively reserved compared with the conventional integration methods. And the attenuation degree for the low frequency acceleration signal can be regulated by the value of the accuracy control parameter to some degree. Therefore, the LFA method achieves the aim for not only eliminating the low frequency noise, but also reserving the effective signal.

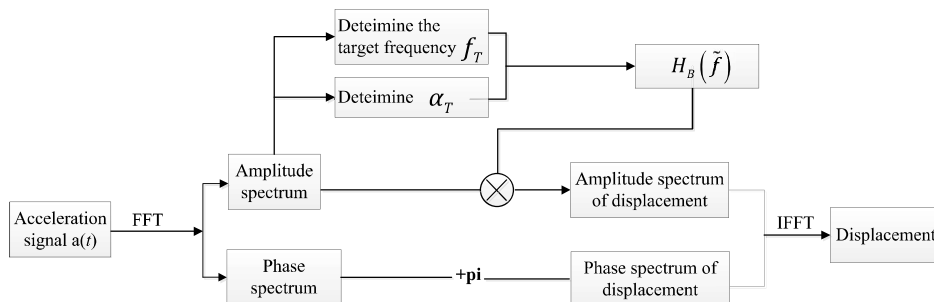


Fig. 7. The flowchart of the LFA integration algorithm.

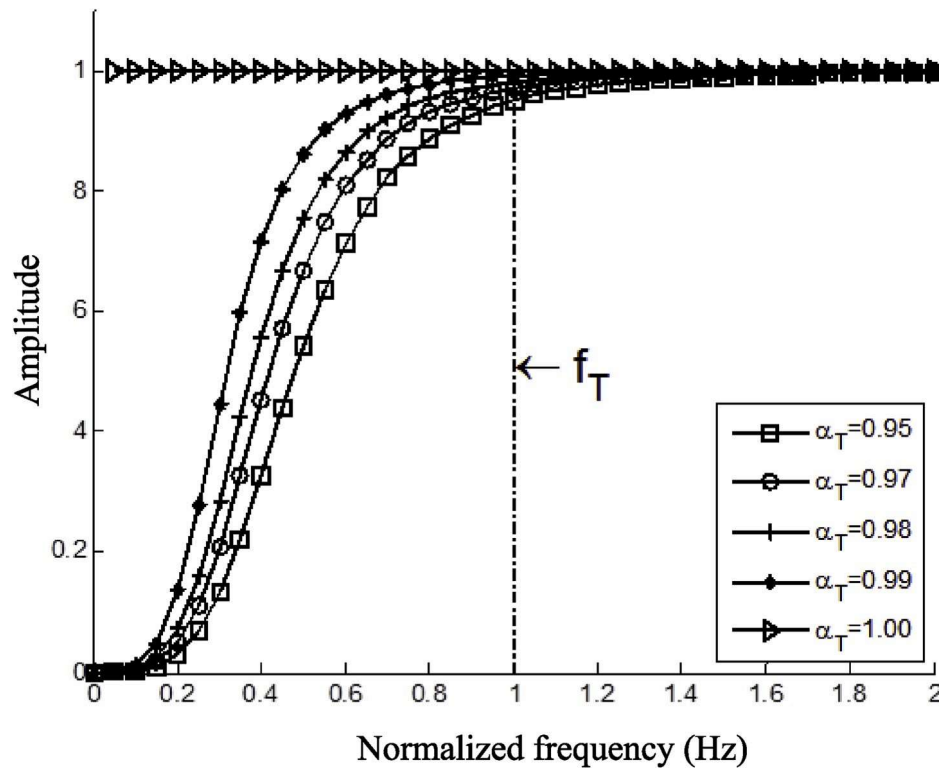


Fig. 8. Amplitude-frequency characteristics of $\hat{a}$ and $\bar{a}$ for the LFA algorithm with different values of ACFs α_T .

3. Comparison on integral accuracy of different integration algorithms

To validate the accuracy of the LFA algorithm and verify its advantage against other conventional integration algorithms, a FE simulation example is implemented using the commercial FE software LS-DYNA. As seen in Fig. 9, there is a one degree freedom transient FE model which consists of two parts: a spring and a reciprocating motion bar. One end of the spring is connected to the reciprocating motion bar, and the other

end is fixed. The other end of the reciprocating motion bar is exerted with a force curve (see Fig. 10) in Z direction and all its six degrees of freedoms are fixed but the translational degree of freedom in the Z direction. The FE simulation is used to calculate transient forced response of the reciprocating motion bar. The corresponding acceleration and displacement curve of the reciprocating motion bar are thus obtained at the same time. FE simulation is in fact based on a nodal-displacement algorithm that the displacements of each node in an element are firstly calculated and then the corresponding velocity and acceleration

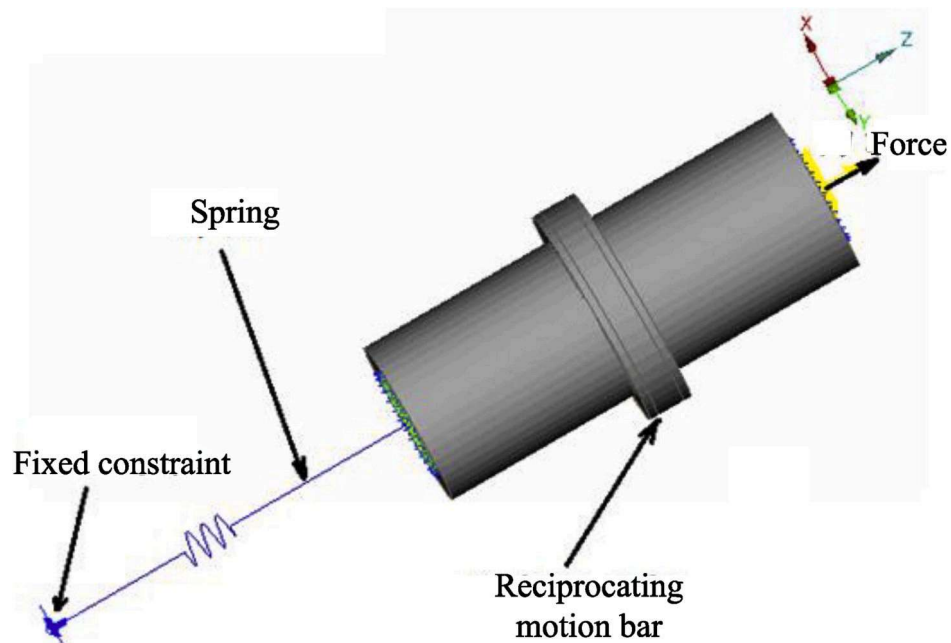


Fig. 9. Finite element model.

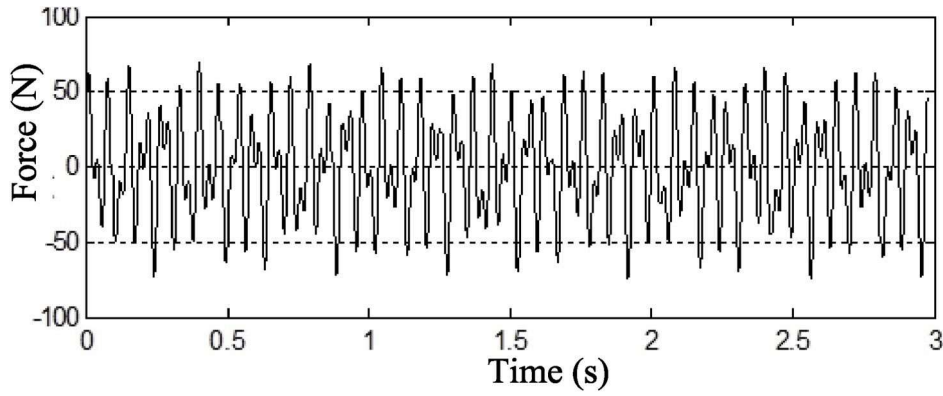


Fig. 10. The load curve of the finite element model.

of the nodes are derived by numerically differentiating the sampled displacement with respect to time in the FE software LS-DYNA. As we all know, acceleration can be accurately obtained from numerical differentiation of displacement with respect to time. Hence the displacements and accelerations obtained from the FE simulation can be regarded as real digital signals obtained from sampling. To test the integral accuracies of the LFA algorithm and the other integration algorithms talked above, the sampled acceleration signal is used to reconstruct displacements by all these algorithms. The reconstructed displacements from all the integration algorithms are compared with the ‘real’ displacement obtained from the FE simulation.

Making Fourier transform for the simulated acceleration and then the amplitude-frequency spectrum is drawn in Fig. 11. As seen from Fig. 11, the base frequency of the acceleration is 6.5 Hz and there are not apparently low-frequency noise signals.

According to the flowchart in Fig. 7, the simulated acceleration is used to reconstruct displacement through the LFA algorithm. The integral result is then compared with the displacement reconstructed by the high-pass filter algorithm, direct integration method, LFCO algorithm and PF algorithm as shown in Fig. 12. The best-fit parameters of each integration algorithm are used here to reconstruct displacements ensuring that each integration algorithm has the best integration accuracy. Values of the target frequency and ACF of the LFA algorithm are set to be 6.5 Hz and 0.98 for integration, respectively. The polynomial of the PF algorithm for fitting the trend error is $0.0030x^2 - 0.8651x + 1.1889$ and the cut-off frequency of the LFCO algorithm is 7/3 Hz. A second order high-pass FIR filter is employed for the filter algorithm and its cut-off frequency is 6 Hz.

Fig. 12(b) and (c) are partial enlarged drawings of the section marked with circles in Fig. 12(a). It is demonstrated in Fig. 12 that almost every integration method has a good accuracy for reconstructing displacements except the direct integration method. But as seen from Fig. 12(b) and (c), we can conclude that the LFA algorithm has the best

integration accuracy compared with other integration methods and the PF algorithm ranks the second.

4. Integral accuracy control based on the LFA algorithm

Like other FDI methods, The LFA algorithm also controls the trend term error through suppressing the low frequency signal. But the attenuation degree for the low frequency acceleration signal can be regulated by the value of the ACF to a certain degree. So how to appropriately select the ACF is of great importance to the reconstructed displacement in engineering application. In this section, the effect of the ACF on integration results will be investigated through a numerical study and a vibration test.

4.1. Error evaluation indices

Three kinds of error evaluation indices including average peak error, average deviation error and absolute value error are introduced to objectively evaluate the difference between the reconstructed displacement waveform and the real one. Schematic of the three error indices are shown in Fig. 13.

The average peak error (Erp) is defined as the average value of the relative error of negative and positive peak values of the time history of the reconstructed displacement $S'(t)$ relative to corresponding values of the real one $S(t)$ [26].

$$Erp = \frac{1}{2} \times \left\{ \left| \frac{\max[S'(t)] - \max[S(t)]}{\max[S(t)]} \right| + \left| \frac{\min[S'(t)] - \min[S(t)]}{\min[S(t)]} \right| \right\} \quad (30)$$

The average deviation error (Err) is defined as the average value of the relative error of the peak value of the displacement time history $S(t) - S'(t)$ relative to that of the real displacement time history $S(t)$ [26].

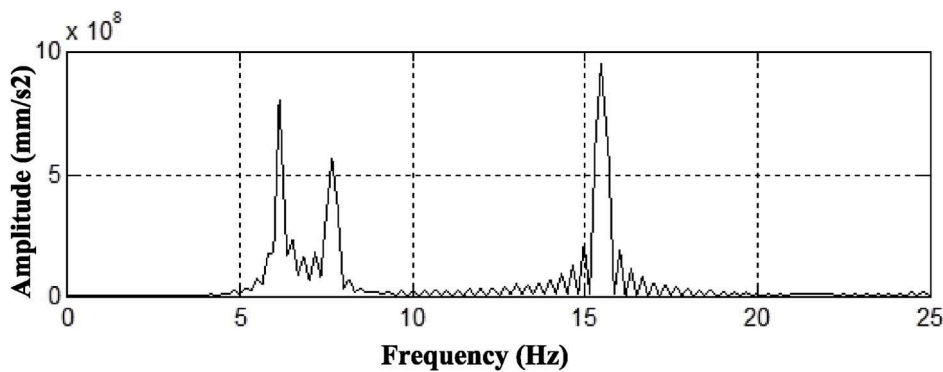


Fig. 11. The amplitude spectrum of acceleration.

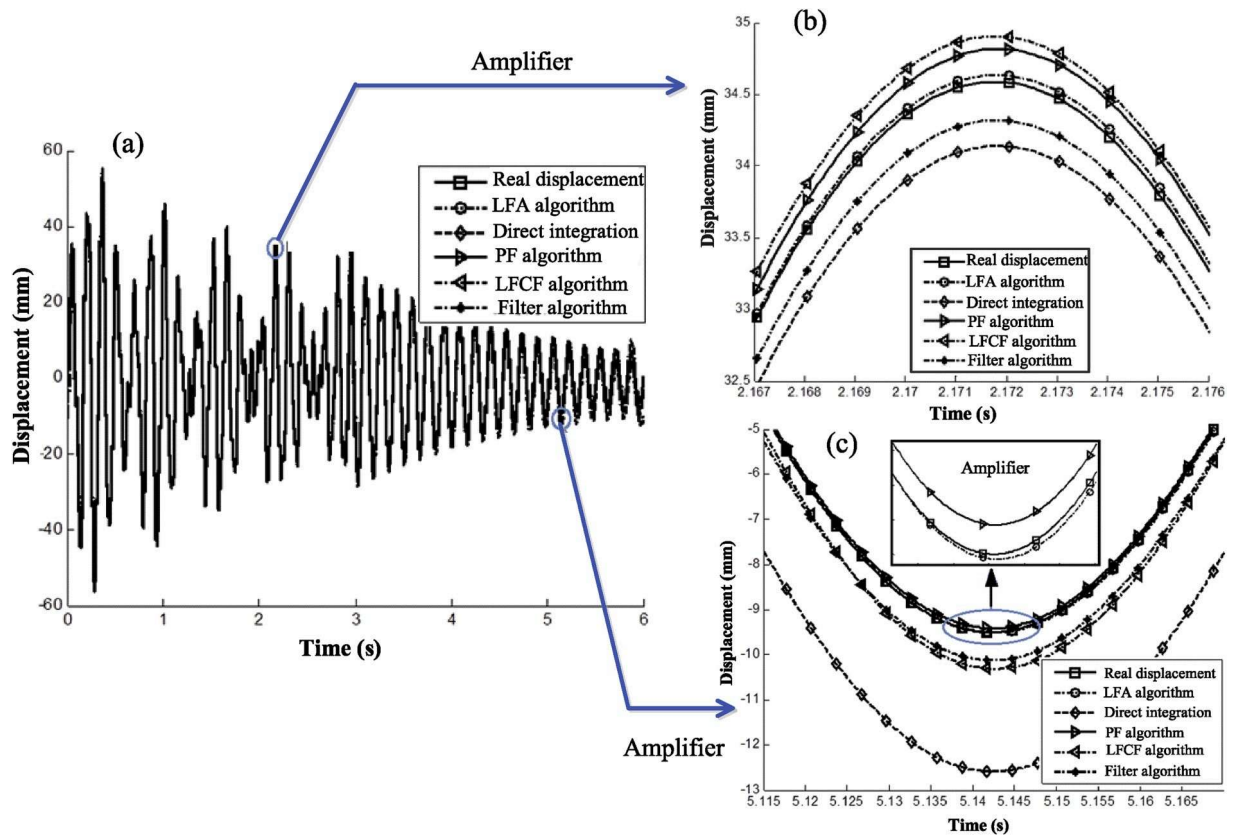


Fig. 12. Comparison with the real displacement and displacements reconstructed by the LFA algorithm, high-pass filter algorithm, direct integration method, LFCO algorithm and PF algorithm from the simulated acceleration, respectively.

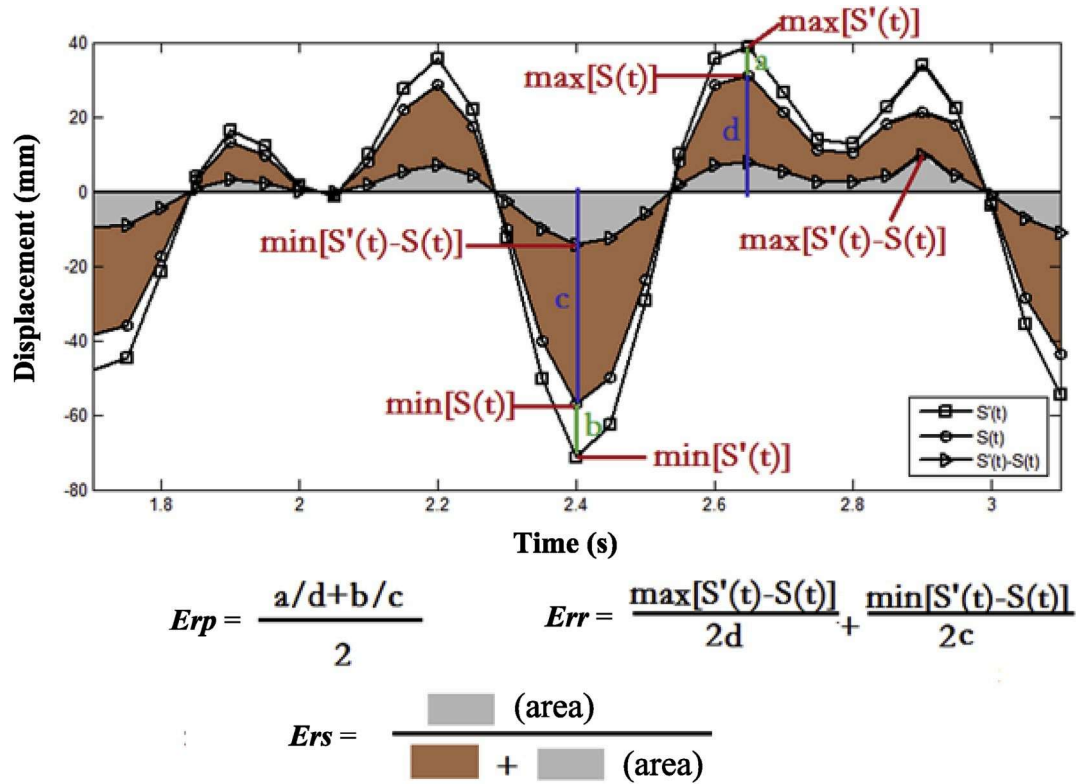


Fig. 13. Schematic of the three error indices.

$$Err = \frac{1}{2} \times \left\{ \left| \frac{\max[S'(t) - S(t)]}{\max[S(t)]} \right| + \left| \frac{\min[S'(t) - S(t)]}{\min[S(t)]} \right| \right\} \quad (31)$$

The absolute value error (*Ers*) is defined as the ratio of $\int_0^t |S'(\tau) - S(\tau)| d\tau$ and $\int_0^t |S(\tau)| d\tau$.

$$Ers = \frac{\int_0^t |S'(\tau) - S(\tau)| d\tau}{\int_0^t |S(\tau)| d\tau} \quad (32)$$

where $\int_0^t |S'(\tau) - S(\tau)| d\tau$ is the area of region surrounded by the time history curve $S'(t) - S(t)$ and the abscissa axis; $\int_0^t |S(\tau)| d\tau$ is the area of region surrounded by the real displacement curve $S(t)$ and the abscissa axis.

4.2. Numerical study

4.2.1. Harmonic superimposed signal

Suppose there is an ideal acceleration signal superimposed by harmonic waveforms as expressed in Eq. (33).

$$a(t) = 8.9 \sin(15.4\pi t) + 35.9 \sin(31\pi t) + 29.3 \sin(56\pi t) \quad (33)$$

If the initial displacement is zero, the real displacement function is derived by double integration of Eq. (33).

$$s(t) = -\frac{8.9}{(15.4\pi)^2} \sin(15.4\pi t) - \frac{35.9}{(31\pi)^2} \sin(31\pi t) - \frac{29.3}{(56\pi)^2} \sin(56\pi t) \quad (34)$$

The acceleration signal of Eq. (33) is sampled with a sampling frequency of 1000 Hz and the total sampling time is 3s. The sampled acceleration waveform is demonstrated in Fig. 14. Through the Fourier transform of the sampled acceleration waveform, the amplitude-frequency spectrum is obtained as shown in Fig. 15.

There are three main frequency components (i.e., 7.7 Hz, 15.5 Hz and 28 Hz) contained in the acceleration signal according to Eq. (33). Hence, truncation error takes place due to the non-integer-period sampling of the acceleration signal as shown in Fig. 15. Since the actual test signal in engineering application often has complex frequency components, the integer-period sampling for all frequency components is nearly impossible within limited sampling time. In other words, truncation error is inevitable in engineering applications.

4.2.2. Selection of the ACF and error evaluation

The minimum principal frequency or the base frequency of the acceleration signal is selected as the target frequency f_T in the LFA algorithm. Hence, in this numerical study, the target frequency f_T is 7.7 Hz. To investigate the effect of the ACF on integration results, different values of the ACF ranging from 0.5 to 0.99 are selected to reconstruct displacement from the sampled acceleration. Fig. 16 shows the amplitude-frequency spectrum of the reconstructed displacement with different levels of the ACF. The attenuation zone of the amplitude in Fig. 16 is amplified as shown in Fig. 17.

As shown in Fig. 17, the attenuation degree for the low frequency signal grows with the decreasing value of the ACF which means that the trend term error would be suppressed greater with a less value of the ACF. However, it doesn't mean that the less the value of the ACF is, the better the integration result will be, since the effective low frequency signals are attenuated at the same time. When the value of the ACF α_T is 0.5, the displacement curve reconstructed through the LFA algorithm is compared with the theoretical curve calculated through Eq. (34) (see Fig. 18). The amplified graph of the displacement curve in region between 1.268s and 1.28s are drawn in Fig. 19. The integration error indices including the *Erp*, *Err* and *Ers* can be calculated from the data in Fig. 18.

To evaluate the influence of the ACF on integration accuracy, the *Erp*, *Err* and *Ers* are respectively calculated at diverse levels of the ACF, and results are listed in Fig. 20. It is seen that all the three kinds of integration error indices always keep at a small level (less than 7%) when the value of the ACF is selected within a wide scope ranging from 0.5 to 0.99. Specifically, when the value of the ACF increases from 0.5 to 0.99, all the corresponding integration error indices including the *Erp*, *Err* and *Ers* decrease first and then grow rapidly.

According to the analysis above, when the LFA algorithm is used to reconstruct displacement, the value of the ACF should be appropriately selected according to the level of low frequency noise. The selection principle is trying to choose a high value of the ACF but should guarantee that obvious trend term errors do not appear and the effective low frequency signal is reserved. Take the numerical study in this section for an example; the highest integration accuracy with errors less than 2% can be achieved when the value of the ACF is between 0.90 and 0.97.

4.3. A vibration test

To further validate the accuracy of the LFA algorithm and the influence of the ACF on integration accuracy in engineering application, a vibration test rig with two limit boards of impact is designed.

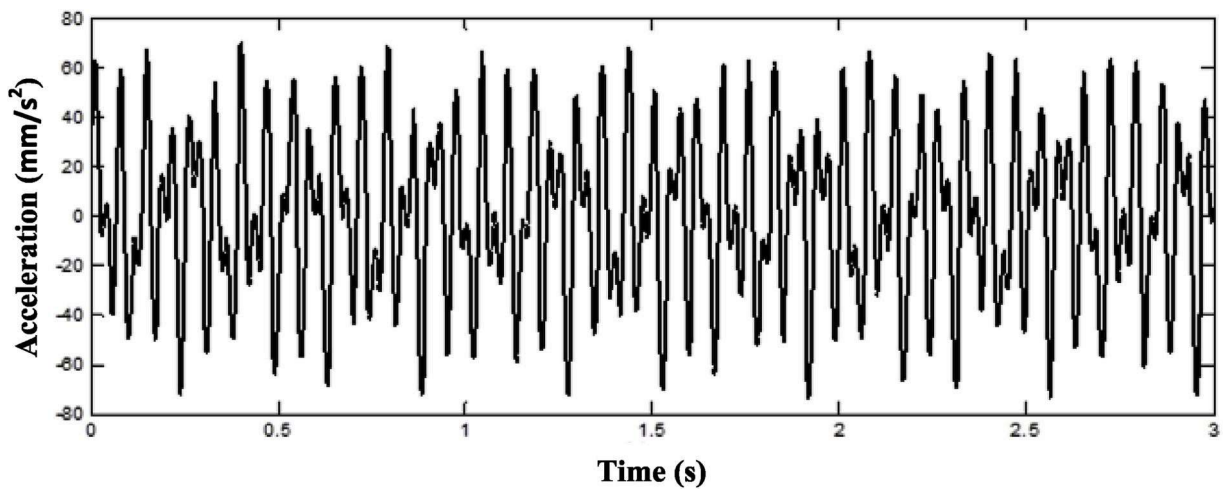


Fig. 14. Time history of sampled acceleration waveform.

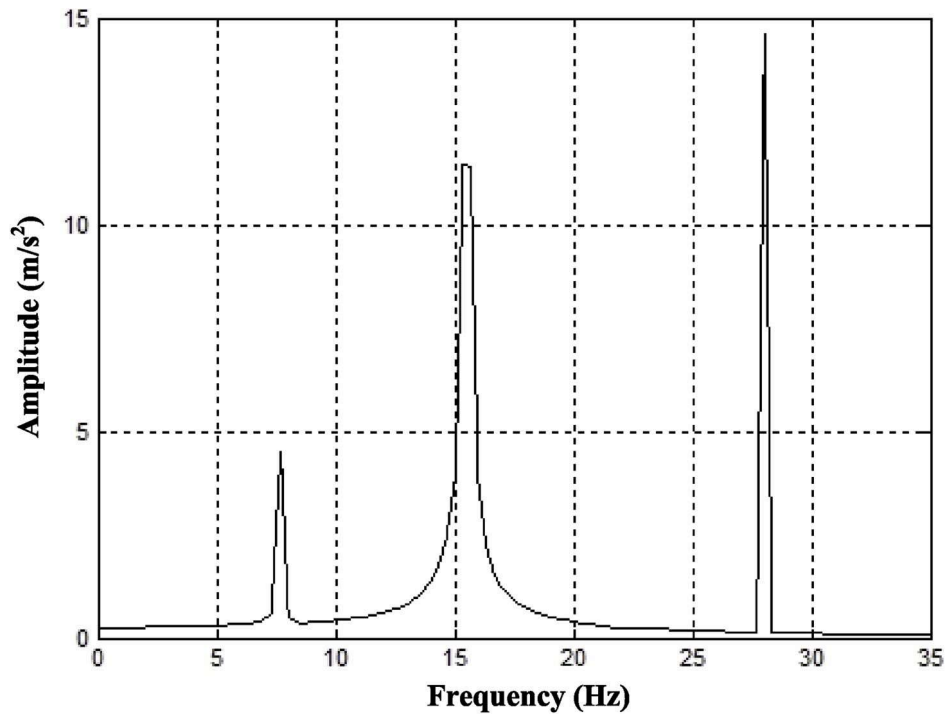


Fig. 15. Amplitude-frequency spectrum for the sampled acceleration waveform.

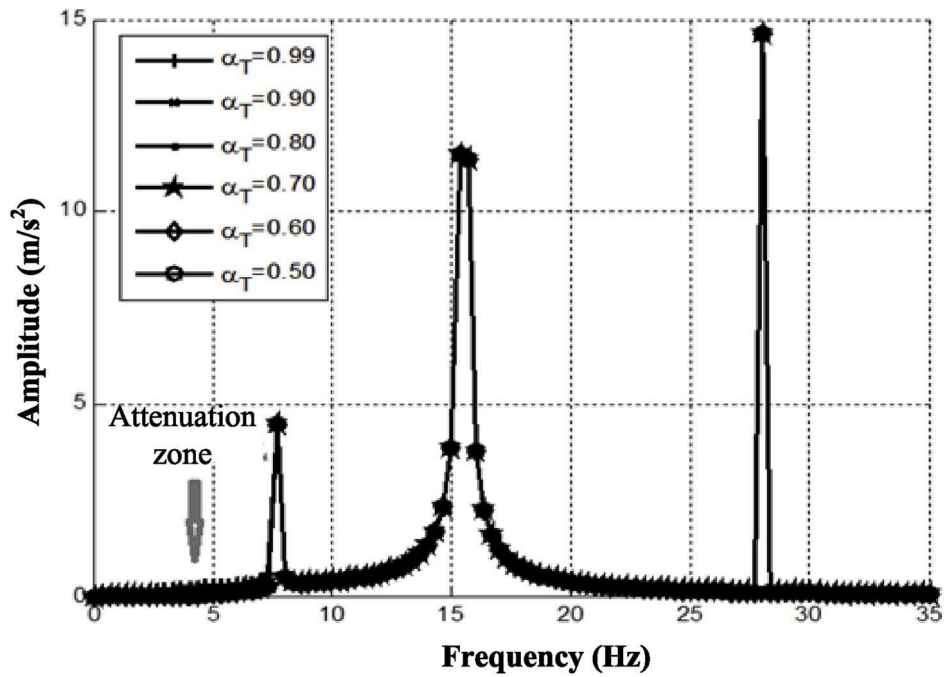


Fig. 16. Amplitude-frequency spectrum of the reconstructed displacement.

4.3.1. Introduction of the vibration test rig

Fig. 21 shows a vibration test rig with two limit boards of impact. This test rig consists of a servo motor, a motorcycle engine crankshaft and a reciprocating bar, etc. The shaft of the servo motor is connected with the engine crankshaft via a shaft coupling. The engine crankshaft is powered by the servo motor and the rotation of the crankshaft is transformed to reciprocating rectilinear movement of the piston through the crank-train within the engine. The piston is connected to the reciprocating motion bar with a spring. There is a shaft shoulder in the middle of the reciprocating bar; and the shaft shoulder, cooperated with two

limit boards fixed to the frame of the engine, acts as a limit function. Hence, the reciprocating motion bar also has similar reciprocating rectilinear movement along its axis. In the reciprocating motion process, cyclic collision occurs between the reciprocating bar and limit boards. Besides, there are two thin layers of rubber sheet on two sides of the shaft shoulder to slow the collision and avoid too large acceleration.

In this test, an accelerometer (PCB-356A25) is employed to measure the acceleration of the reciprocating bar and the sampling frequency is 25.6 KHz. Besides, a laser displacement sensor is used to measure the corresponding displacement at the same time and its sampling frequency

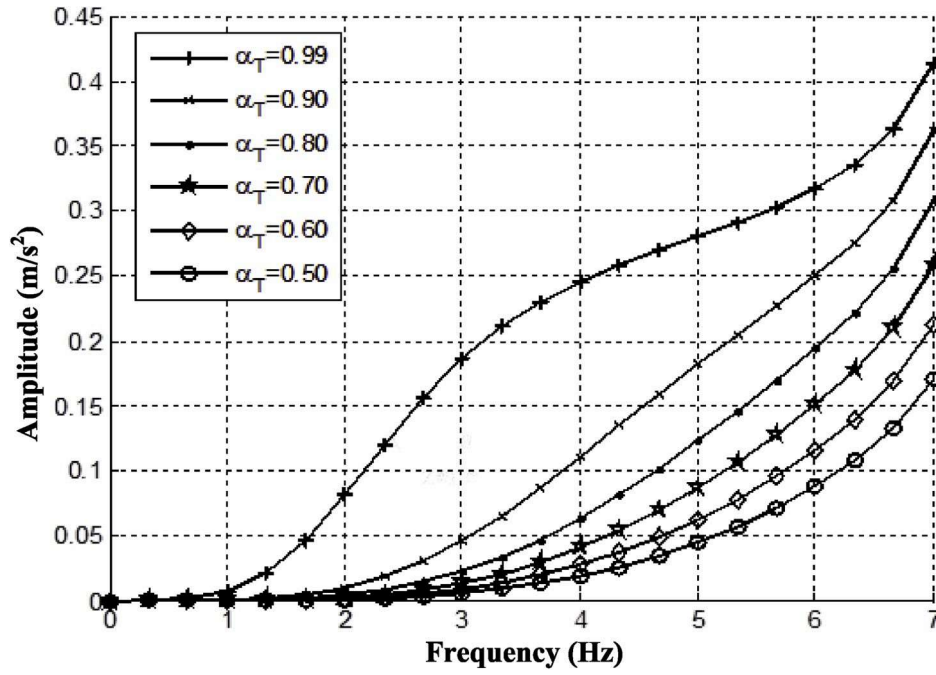
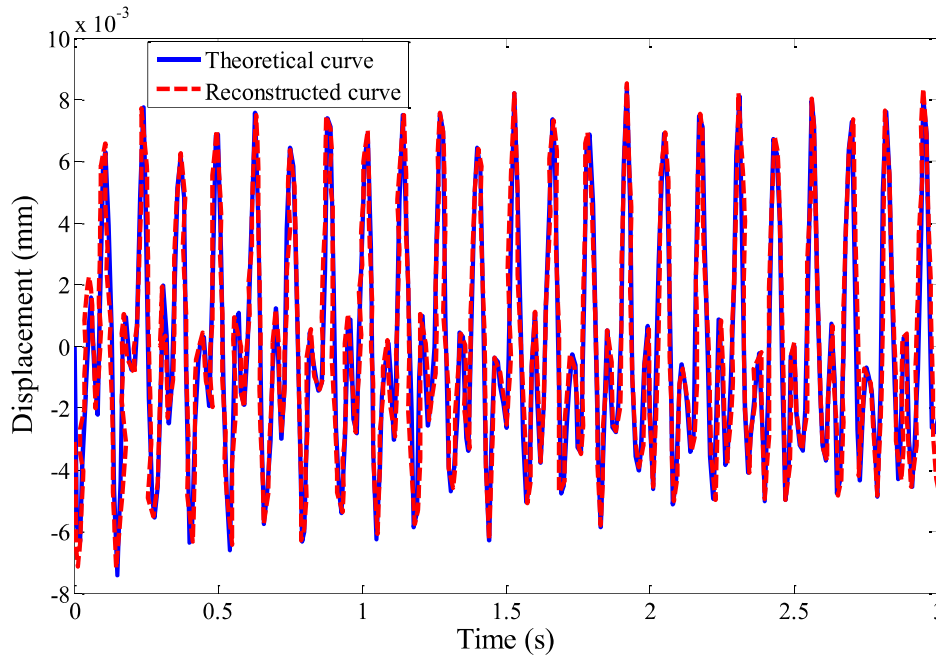


Fig. 17. Amplified graph for the attenuation zone.

Fig. 18. Comparison of theoretical and reconstructed displacement curves at $\alpha_T = 0.5$.

is 200 Hz. The measured acceleration will be used to reconstruct displacements by the LFA algorithm, and the reconstructed displacements will be compared with the measured displacement to test the performance of the LFA algorithm in integration accuracy with different values of ACF.

4.3.2. Selection strategy of the ACF and error evaluation

When the motor's rotation velocity n is 200 rpm, the measured acceleration and displacement curves of the reciprocating bar are drawn in Figs. 22 and 23, respectively.

The LFA method is also used to reconstruct displacement based on the measured acceleration. When the rotation velocity is 200 rpm, the

collision frequency is 3.33 Hz which is selected as the value of the target frequency. The reconstructed displacement through the LFA algorithm is also presented in Fig. 23, when the value of the ACF is 0.94. The amplified graph of the displacement curve in the region between 1.8s and 2.3s in Fig. 23 is drawn in Fig. 24. From the comparison between the measured and reconstructed displacements, one can see that there are great differences between them at the beginning time span from 0 to 0.5s and at the ending time span from 9.5s to 10s, while good agreement is achieved at the middle time span from 0.5s to 9.5s.

To further evaluate the influence of the ACF on integration results in engineering application, different values of the ACF is selected when the LFA algorithm is used to reconstruct displacement. Likewise, the three

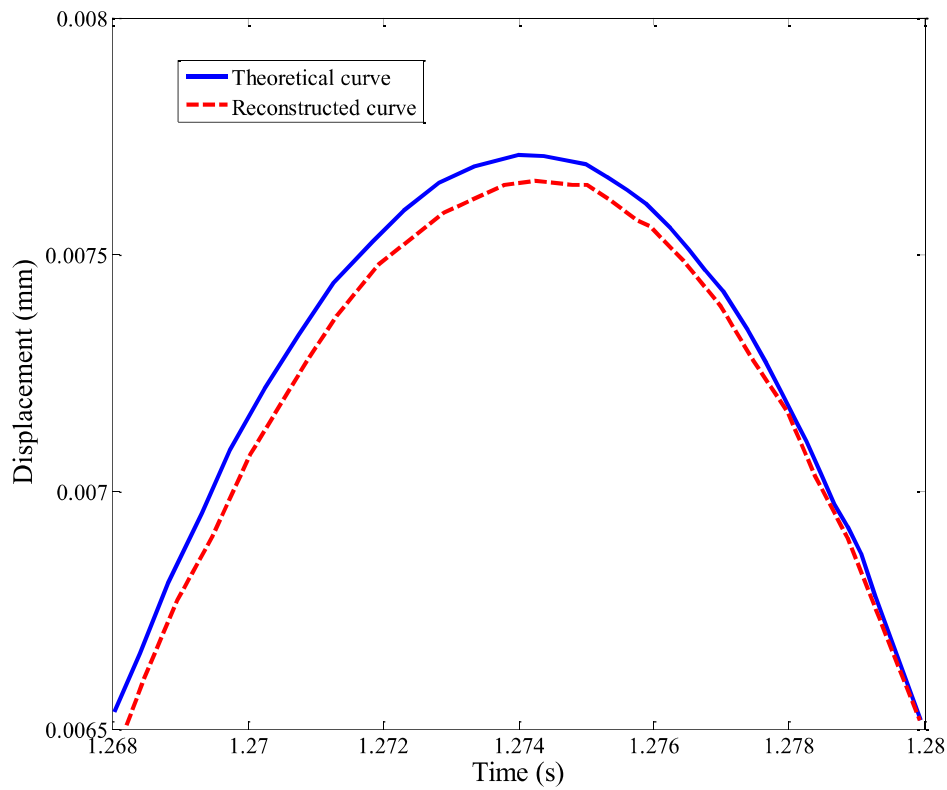


Fig. 19. Details of the comparison of theoretical and reconstructed displacement curves in Fig. 18 from 1.268s to 1.28s.

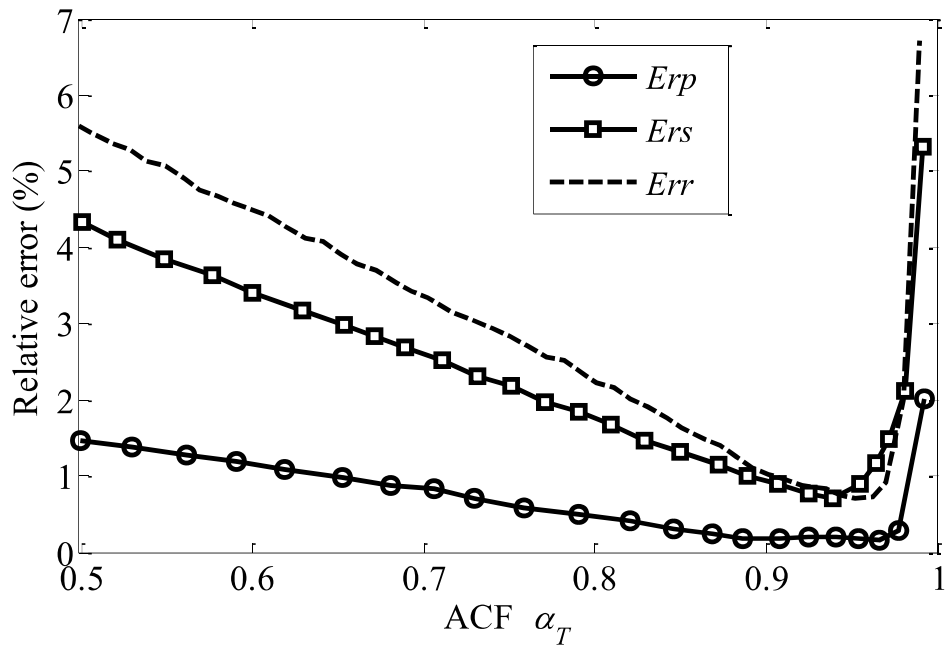


Fig. 20. Integral errors for the LFA algorithm with diverse levels of ACF from 0.5 to 1.

kinds of error evaluation indices proposed are calculated at different levels of the ACF and the statistical results are listed in Fig. 25. As seen from Fig. 25, when the value of the ACF is set between 0.900 and 0.999, all the three integration errors keep pretty small and are less than 4%. What's more, all the three kinds of integration error indices have small sensitivities to the value of ACF when its value is larger than 0.9. The integration error slowly decreases first and then increases fast with increasing value of the ACF. When the value of the ACF is selected

between 0.98 - 0.99, a best integration result can be obtained.

Through the numerical study and vibration test research, we can conclude that the value of the ACF has a great influence on the integration accuracy. And the value of the ACF is suggested to be selected between 0.90 and 0.98 to achieve a good integration result in engineering application.

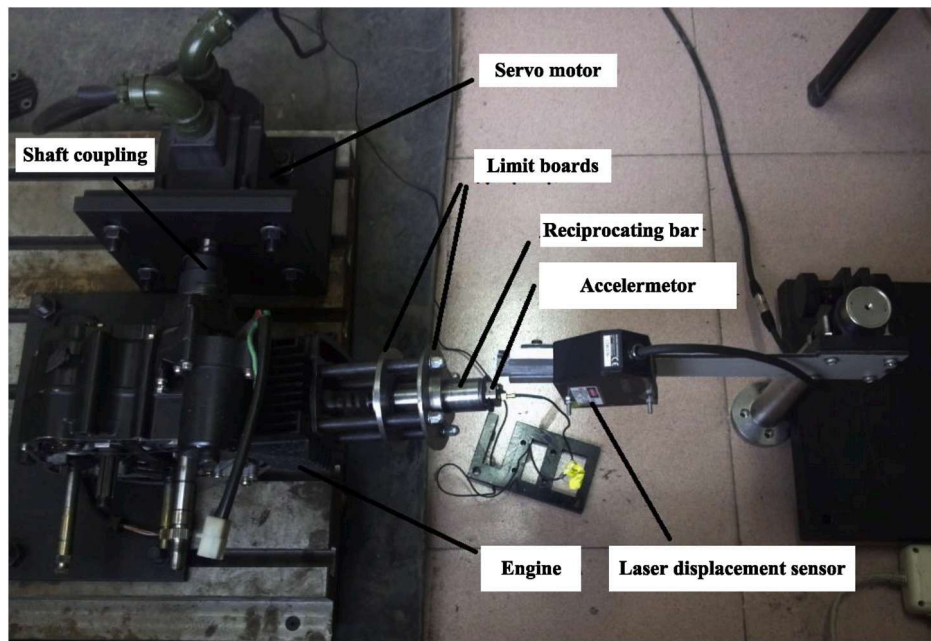
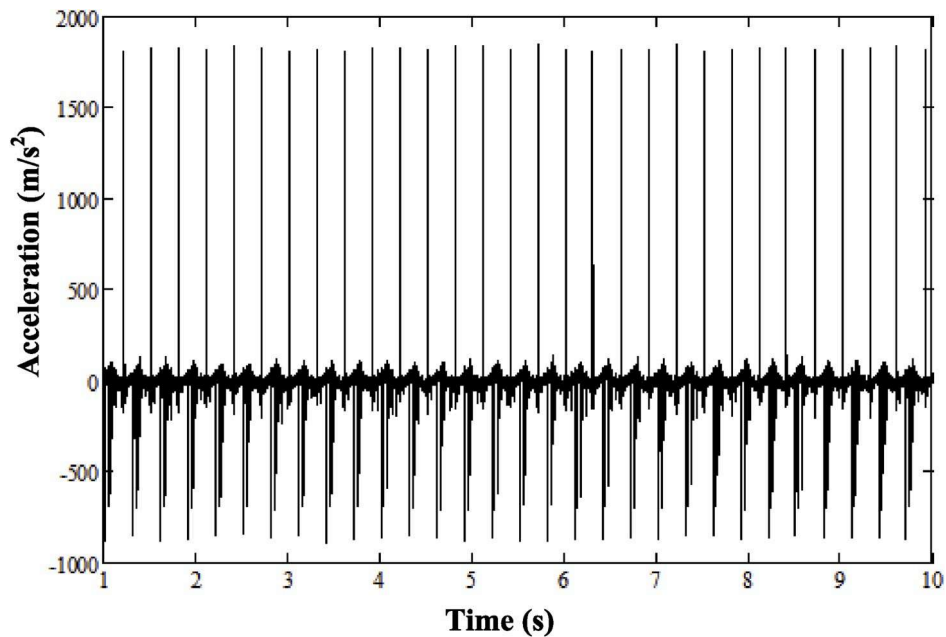


Fig. 21. Vibration test rig.

Fig. 22. Measured acceleration curve at $n = 200$ rpm.

5. Conclusion

The LFA algorithm is used in this work to directly integrate the acceleration signal in frequency domain and an ACF is employed to assure the integration accuracy. Through a FE simulation study, it is validated that the LFA algorithm has more advantages in integration accuracy compared with other conventional integration algorithms. At the same time, a numerical study and a vibration test are designed to investigate the influence of the ACF in the LFA algorithm on integration accuracy, and three error evaluation indices are proposed to evaluate the integration accuracy. Several conclusions are drawn from the comparison.

- 1) The LFA algorithm rapidly suppresses the frequency far below the target frequency, but the signals with frequencies around or above the target frequency are effectively reserved compared with the conventional integration methods. Therefore, the LFA method achieves the aim for not only eliminating the low-frequency noise but also retaining the effective signal.
- 2) The attenuation degree for the low frequency acceleration signal can be regulated by the value of the ACF. The less the value of the ACF, the greater the attenuation degree.
- 3) The numerical study shows that all the three kinds of integration errors always keep at a small level (less than 7%) when the value of the ACF is selected within a wide scope ranging from 0.5 to 0.99. The highest integration accuracy with an error less than 2% can be

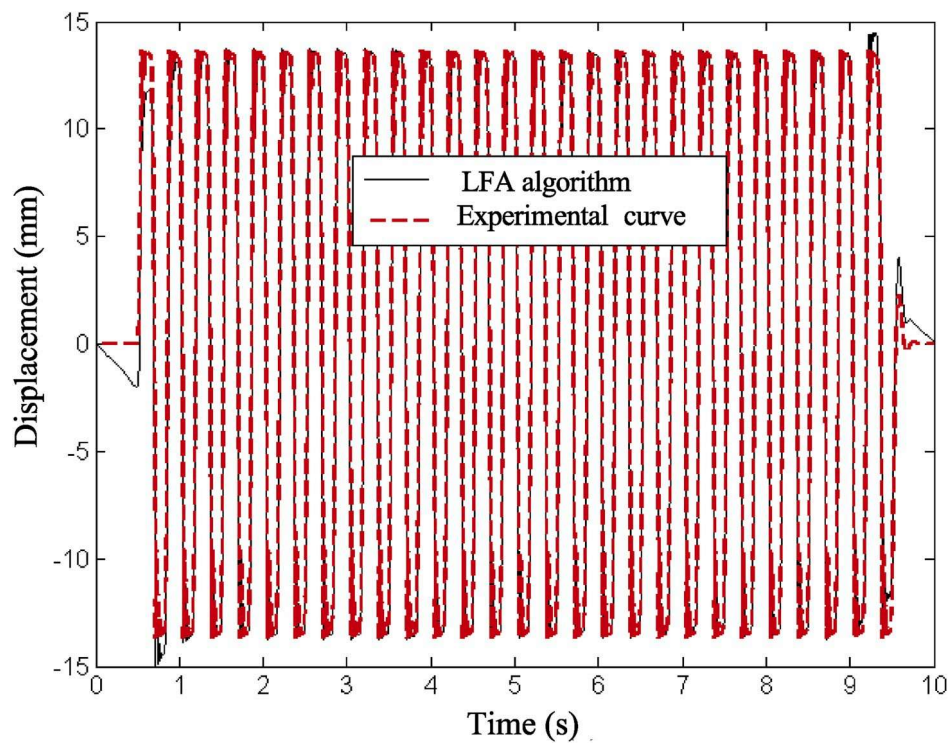


Fig. 23. Comparison between the experimental displacement curve and the reconstructed displacement curve integrated from the measured acceleration curve in Fig. 22 using the LFA algorithm.

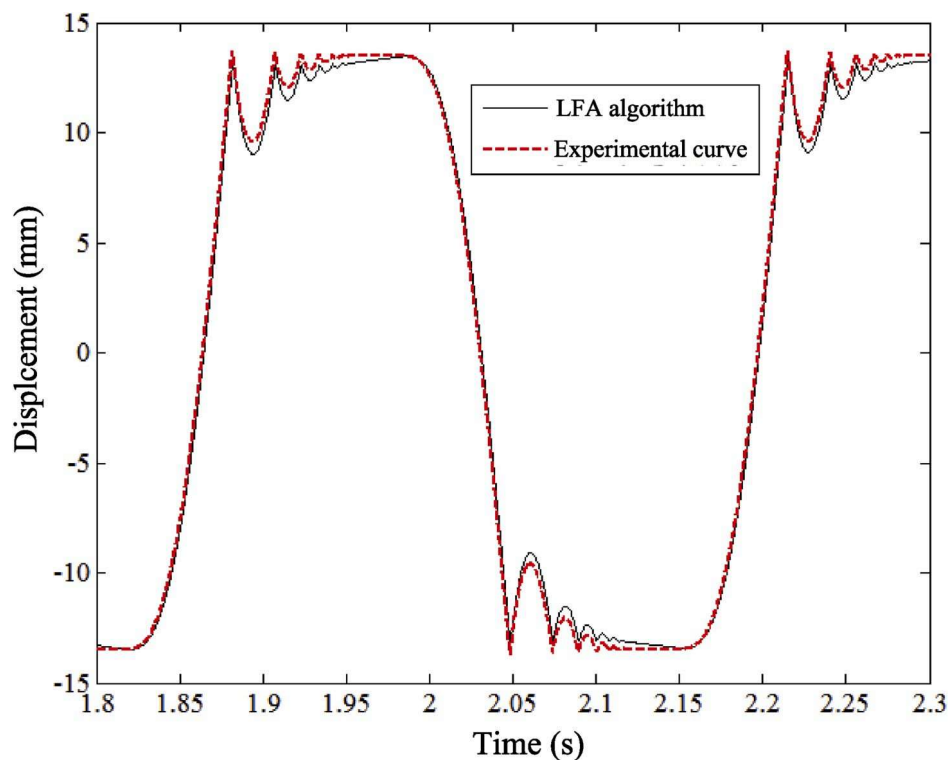


Fig. 24. Details of the displacement curves in Fig. 23 at $n = 200$ rpm when the value of the ACF is 0.94.

achieved when the value of the ACF is located in the range $[0.9, 0.97]$.

- 4) The vibration test demonstrates that the integration errors keep pretty small which are less than 4%, when the value of the ACF is set between 0.900 and 0.999. Still, all the three kinds of integration

error indices are a little sensitive to the value of the ACF, and thus the value of the ACF is easy to be determined.

All in all, selection of the value of the ACF should take full consideration of the level of low frequency noises, and it is better to select a

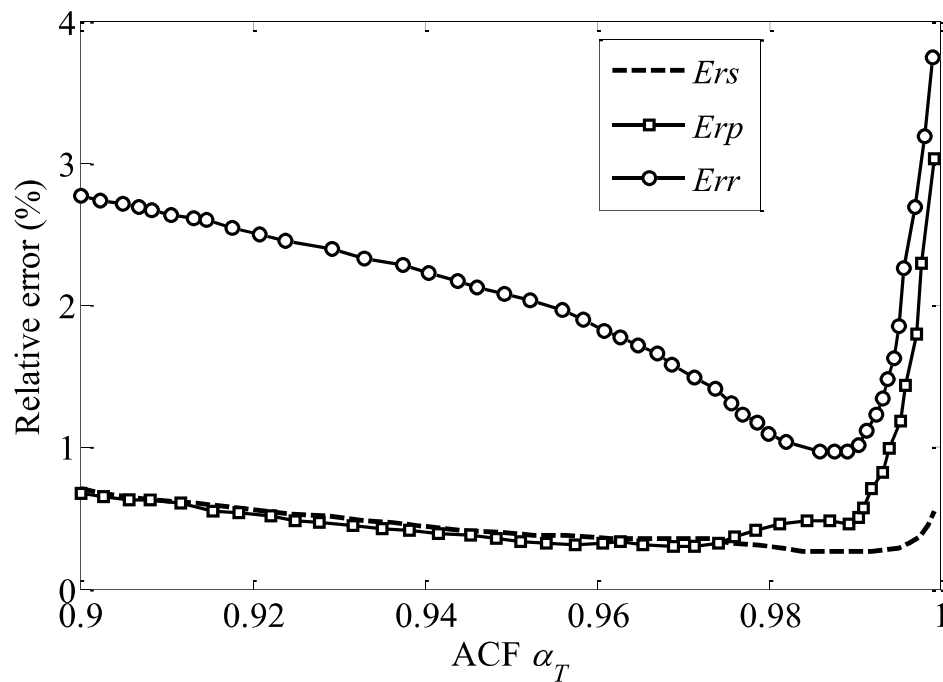


Fig. 25. Integral errors for the LFA algorithm.

large value to ensure a good effect for eliminating the trend term error or drift error. According to the research, the value of the ACF is suggested to be within the range from 0.90 to 0.98 that good integration accuracies can be obtained in engineering applications. The FE simulation, the numerical study and the vibration test used in this work have different vibration signals in nature (periodic or non-periodic, harmonic or non-harmonic). These three examples are typical and representative. Therefore, the LFA algorithm has certain generality and is applicable for many real occasions.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and it is approved by all authors for publication. All funding sources supporting the work and the institutional or corporate affiliations of the authors have been acknowledged in this work. I would like to declare on behalf of my co-authors that the work has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

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